Four-Factor Analysis of UK Asset Returns

(Group work)

1. Introduction

This study conducts an empirical analysis of ten equity portfolios, developed using empirical data provided to us, following Fama–French's three-factor model and Carhart's (1997) four-factor model, which adds the momentum factor (UMD/MOM) to Fama–French's original three factors. The raw monthly portfolio returns are labelled **S1–S10** in the source data; throughout the rest of this report, **XS1–XS10** denotes their excess returns over the risk-free rate, which is the series actually used in every regression, table and figure that follows.

The objective of this analysis is first and foremost to estimate the risk factors (Beta) for all ten equity portfolios, using the four-factor model, after calculating all excess returns and interpreting their results. We then verified the cross-sectional relationships, identifying the relationships between average returns and estimated betas. In addition, we assessed the impact of outliers using the "Winsorising" technique, comparing the original data with the "cleaned" data to understand whether they had an impact on our analysis. Finally, we carried out a general comparison of the overall data and identified all the factors that were significant in determining returns.

1.1 Data and Methodology

The dataset provided consists of a data matrix whose data is calculated on a monthly basis from 1980 to 2010. The variables are classified as follows:

- **S1–S10:** Monthly returns of 10 equity portfolios
- **RF:** Risk-free rate (1-month Treasury Bill)
- **RMRF:** Market excess return (rm − rf)
- **SMB:** Size factor (Small Minus Big)
- **HML:** Value factor (High Minus Low book-to-market)
- **UMD:** Momentum factor (Up Minus Down)

Excess Returns

The following formula is used to calculate Excess Returns:

$$XS_i = S_i - R_f$$

Where:

- XS_i = Excess return of portfolio i
- S_i = Total return of portfolio i
- R_f = Risk-free rate

1.3 Time Series Regressions

$$XS_{i,t} = \alpha_i + b_{i,RMRF} \cdot RMRF_t + b_{i,SMB} \cdot SMB_t + b_{i,HML} \cdot HML_t + b_{i,UMD} \cdot UMD_t + \varepsilon_{i,t}$$

Where α is the part of the excess return unexplained by the factors, and $\varepsilon_{i,t}$ is the error term. Each coefficient b measures the portfolio's sensitivity to the corresponding factor.

1.4 Cross-Sectional Analysis

The following formulas were used to perform the cross-sectional analysis, where the dependent variable is each portfolio's mean excess return and the regressors are the betas estimated in the time-series step:

(1) Mean Return = $\gamma_o + \gamma_1 \beta RMRF + \gamma_2 \beta SMB$

(2) Mean Return = $\gamma_o + \gamma_1 \beta RMRF + \gamma_2 \beta HML$

(3) Mean Return = $\gamma_o + \gamma_1 \beta RMRF + \gamma_2 \beta UMD$

1.5 Statistical Tests and Diagnostic Checks

To make sure our regression results were reliable, we ran several basic checks on the data and the residuals. Because financial returns usually do not follow a normal distribution, we used the Jarque–Bera and Shapiro–Wilk tests to see whether the data was normally distributed. We also applied the Breusch–Pagan test to check for heteroscedasticity and the Durbin–Watson statistic to see if there was any autocorrelation in the residuals.

To understand the results, we looked at the t-statistics and p-values of each coefficient, following the usual convention: significant when $p < 0.05$, very significant when $p < 0.01$, and highly significant when $p < 0.001$. Finally, we used the Adjusted R-squared to see how well the models explained the variation in returns. In the cross-sectional regressions, we considered an R^2 above 0.50 to indicate a fairly strong level of explanatory power, especially given the small number of portfolios.

2. Descriptive Statistical Analysis

This section provides a preliminary analysis of the ten UK equity portfolios (XS1–XS10), tracked monthly from 1980 to 2010. To facilitate a clear comparison of market behavior, the portfolios are categorized into three distinct groups based on market capitalization: Small-cap (XS1–XS3), Medium-cap (XS4–XS7), and Large-cap (XS8–XS10).

Before estimating the risk factors using the Fama–French and Carhart models, we examine the fundamental statistical properties of the data—including average returns, volatility, skewness, and kurtosis. These initial diagnostics allow us to observe how risk and return characteristics vary across firm sizes and identify potential deviations from normality.

2.1 Time Series and Portfolio Overview

As illustrated in Figure 1, the portfolios exhibit distinct sensitivities to major macroeconomic and financial shocks over the 30-year horizon. Consistent with expectations, volatility surges across all size categories during periods of market uncertainty.

During the early 1980s recession, an environment characterized by high inflation and contractionary monetary policy resulted in significant return fluctuations. On Black Monday (1987), the largest portfolios (XS6–XS10) experienced the most severe drawdowns, a phenomenon often attributed to the liquidity preference hypothesis, where investors prioritize selling highly liquid large-cap stocks during panic selling. Conversely, during the Dot-com bubble burst (1999–2001), medium-sized portfolios (XS4–XS7) were disproportionately affected, potentially due to higher sectoral exposure to technology firms. In the 2007–2008 financial crisis, volatility intensified again; however, medium portfolios suffered the steepest declines, while small (XS1) and large (XS10) portfolios demonstrated relative resilience.

Overall, the observed patterns largely align with standard asset-pricing theories: while small firms generally offer higher risk premiums, their sensitivity to volatility is acute. These time-series findings underscore the critical role of firm size in determining portfolio behavior across varying economic cycles.

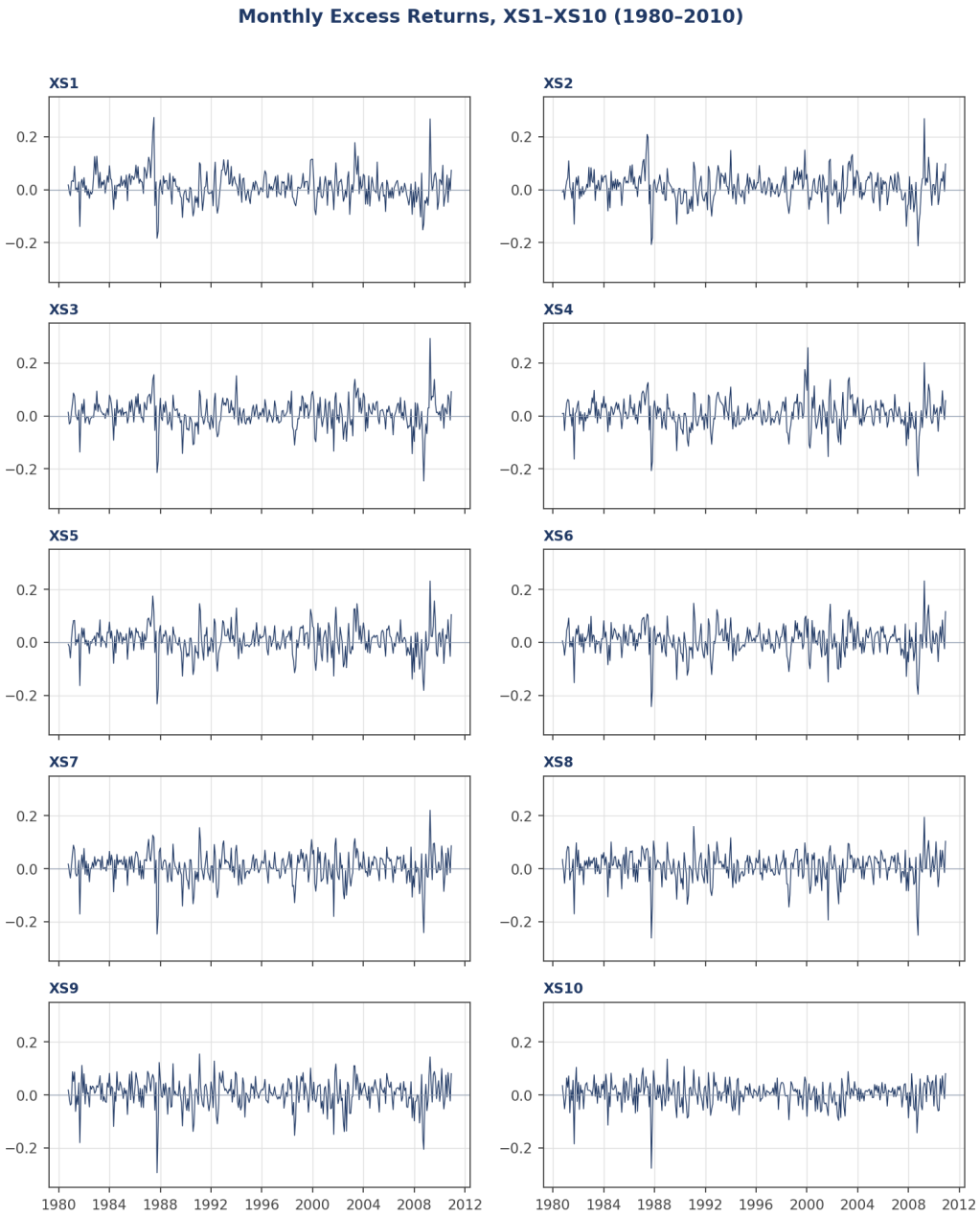


Figure 1. Monthly excess returns, XS1–XS10 (1980–2010)

2.2 Descriptive Analysis

As presented in Table 1, the summary statistics reveal a mean excess return ranging from 0.0109 for the smallest portfolio (XS1) to 0.0050 for the largest (XS10). This pattern is consistent with the well-documented 'size effect,' where smaller firms tend to command a higher risk premium.

Regarding risk metrics, standard deviations cluster between 0.05 and 0.055, indicating elevated volatility for small and medium portfolios relative to large-cap firms. Furthermore, the distributional characteristics suggest significant deviations from normality. Skewness ranges from −1.04 to +0.48, with the majority of portfolios exhibiting negative skewness. Kurtosis values fall between 5.4 and 7.3, pointing to leptokurtic behavior (fat tails). These findings—negative skewness and excess kurtosis—align with the stylized facts of financial asset returns, implying that extreme market movements are more frequent than a normal distribution would predict.

Variable	Mean	Median	SD	Min	Max	Skewness	Kurtosis
XS1	0.0109	0.0097	0.0541	-0.1836	0.2725	0.484	6.435
XS2	0.0105	0.0113	0.0553	-0.2127	0.2682	-0.039	5.949
XS3	0.0087	0.0107	0.0537	-0.2459	0.2928	-0.269	7.071
XS4	0.0079	0.0093	0.0555	-0.2268	0.2578	-0.195	6.018
XS5	0.0072	0.0101	0.0549	-0.2328	0.2314	-0.293	5.358
XS6	0.0076	0.0115	0.0546	-0.2431	0.2318	-0.514	5.603
XS7	0.0067	0.0102	0.0545	-0.2474	0.2206	-0.771	6.213
XS8	0.0066	0.0137	0.0545	-0.2624	0.1942	-1.016	6.564
XS9	0.0067	0.0139	0.0553	-0.2938	0.1548	-0.933	6.053
XS10	0.0051	0.0098	0.0455	-0.277	0.1355	-1.037	7.318

Table 1. Descriptive statistics of excess returns, XS1–XS10

In the histograms with density curve for each XS shown in Figure 2, it can be seen that the distributions deviate from a perfect normal distribution, with heavier tails and, in some cases, asymmetries. All this is consistent with the calculated skewness and kurtosis. These data gave rise to the need for normality tests (Shapiro–Wilk, Anderson–Darling, Jarque–Bera). The statistical tests confirm that the data are not normally distributed, providing overwhelming evidence of non-normality across the sample.

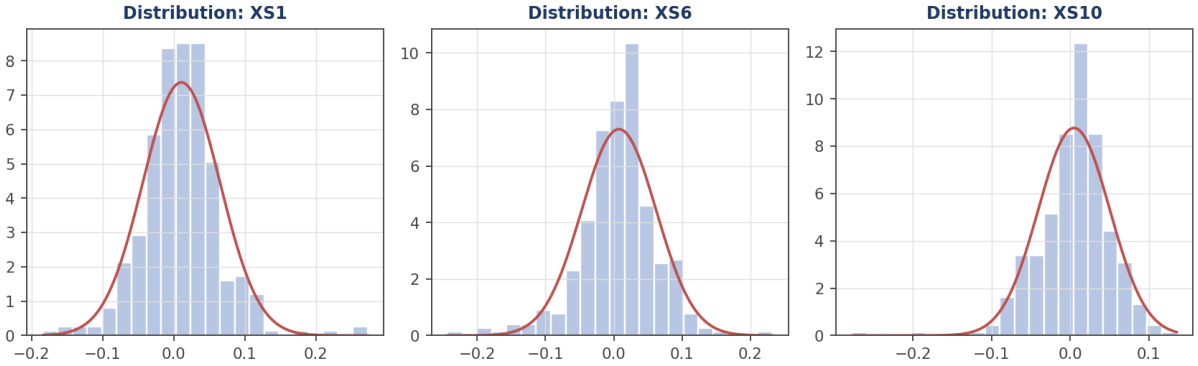


Figure 2. Distribution of excess returns for XS1, XS6 and XS10, with fitted normal density

2.3 Visualization and Correlation Analysis

2.3.1 Boxplots

Figure 3 presents the boxplots of the raw excess returns. In the raw data, small portfolios (XS1–XS3) show more positive outliers, while large portfolios (XS8–XS10) display more negative ones. This suggests that small firms occasionally generate very high returns, whereas large firms are more vulnerable to sharp drops in weak market conditions.



Figure 3. Boxplot of raw excess returns

2.3.2 Correlation analysis

As shown in Figure 4, all correlations fall between 0.51 and 0.95, indicating that the portfolios generally move in the same direction when market conditions change. The relationship is stronger among small and medium portfolios (XS1–XS6), while the correlations are relatively lower between small and large portfolios (especially XS1–XS3 with XS8–XS10). This shows that portfolios with similar firm sizes behave more similarly, whereas those with different sizes move more independently. Overall, the results suggest that all portfolios are influenced by broad market movements (RMRF), but size still plays a role. The differences between small and large portfolios reflect the size effect captured by the SMB factor.

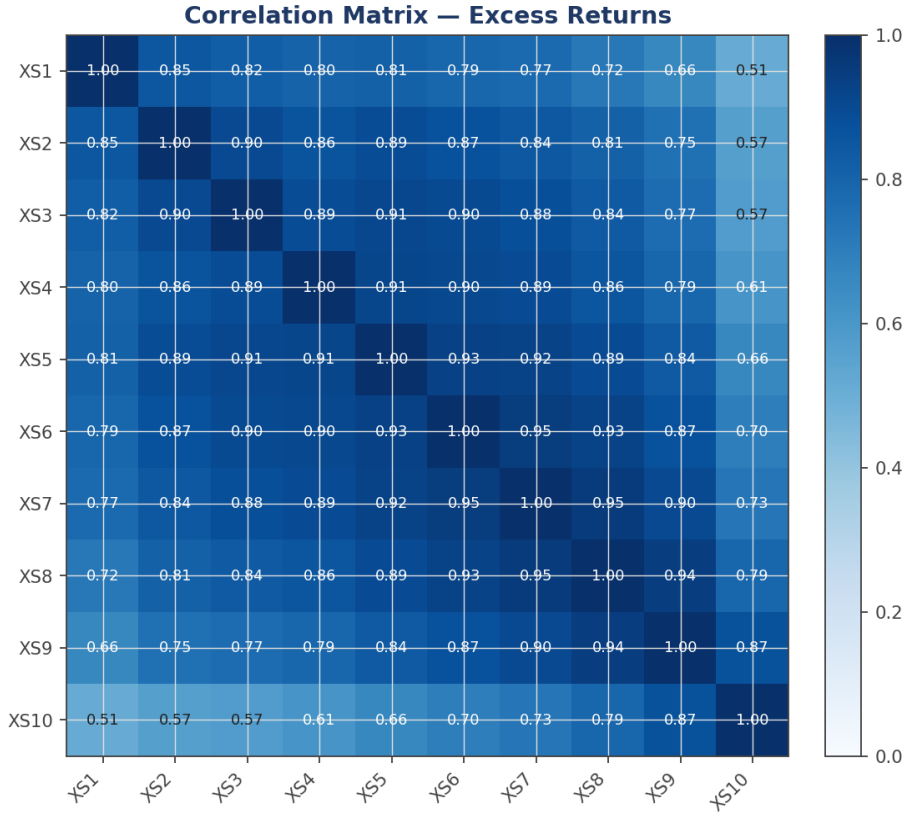


Figure 4. Correlation matrix of excess returns

The descriptive and visual analyses show clear differences in average returns, risk levels, and correlations between small, medium, and large portfolios. These results are consistent with the general idea of the Fama–French model, suggesting that firm size and market exposure may influence how returns change over time.

3. Regression Analysis of Portfolio Excess Returns

3.1 Overview

The core of the analysis consists of time-series regressions. This model is estimated for each XS1–XS10, saving coefficients, t-statistics, and diagnostic graphs. The results show that market betas increase as firm size increases. Specifically, market betas range from about 0.67 for XS1 to around 1.08 for XS10. The t-statistics of these coefficients are extremely high (often above 20–30), confirming strong significance. In contrast, the explanatory power of the model increases with size, with R^2 values ranging from about 0.60 (XS1) to 0.96 (XS10).

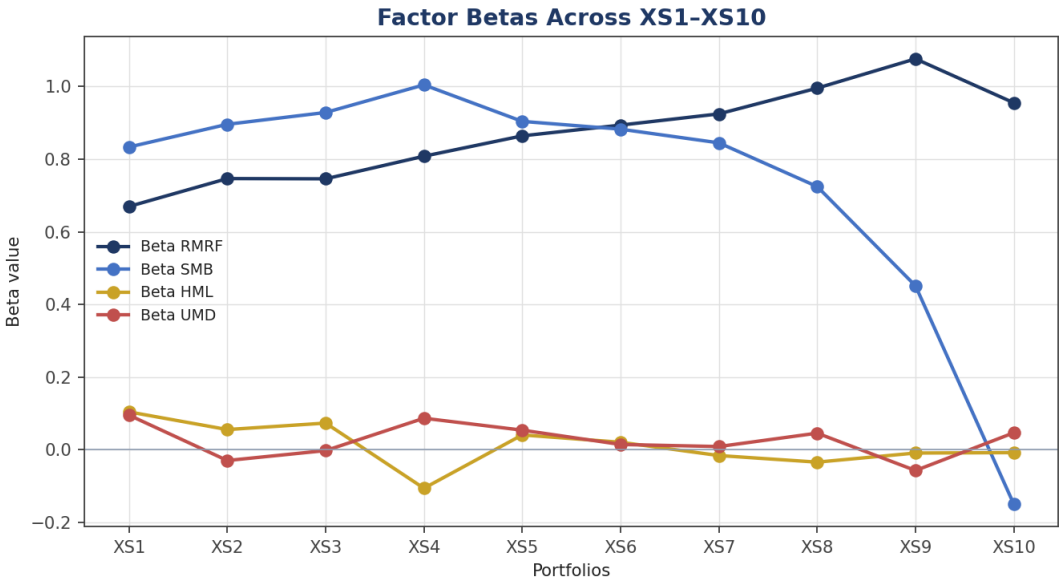


Figure 5. Factor betas across XS1–XS10

Market beta (RMRF): there is an upward trend; portfolios XS1 to XS9 show a gradual and steady increase in market exposure from 0.67 to 1.08. The last one (XS10), on the other hand, falls slightly to 0.96.

SMB Beta (Size): portfolios XS1–XS5 have a very strong and positive exposure to the size factor (from 0.83 to 1.00), indicating that they are mainly composed of small caps. Exposure drops dramatically to become negative for XS10 (−0.15), which behaves like a large cap.

Beta HML & UMD: finally, the coefficients of these two categories are generally low (close to zero), indicating that these portfolios do not have strong systematic exposure to the Value and/or Momentum factors.

XS	Alpha	β RMRF	β SMB	β HML	β UMD
XS1	0.0054	0.6697	0.8334	0.1042	0.0952
XS2	0.0057	0.7465	0.8959	0.0555	-0.0297
XS3	0.0035	0.7459	0.9285	0.0733	-0.0024
XS4	0.0024	0.8081	1.0046	-0.1061	0.0867
XS5	0.0012	0.8641	0.904	0.0407	0.0538
XS6	0.0018	0.8937	0.8826	0.0205	0.0144
XS7	0.001	0.9246	0.8449	-0.016	0.0087
XS8	0.0005	0.9955	0.7245	-0.0342	0.0456
XS9	0.0012	1.0762	0.4511	-0.0091	-0.0573
XS10	0	0.9551	-0.15	-0.0081	0.0466

Table 2. Estimated factor betas from the time-series regressions

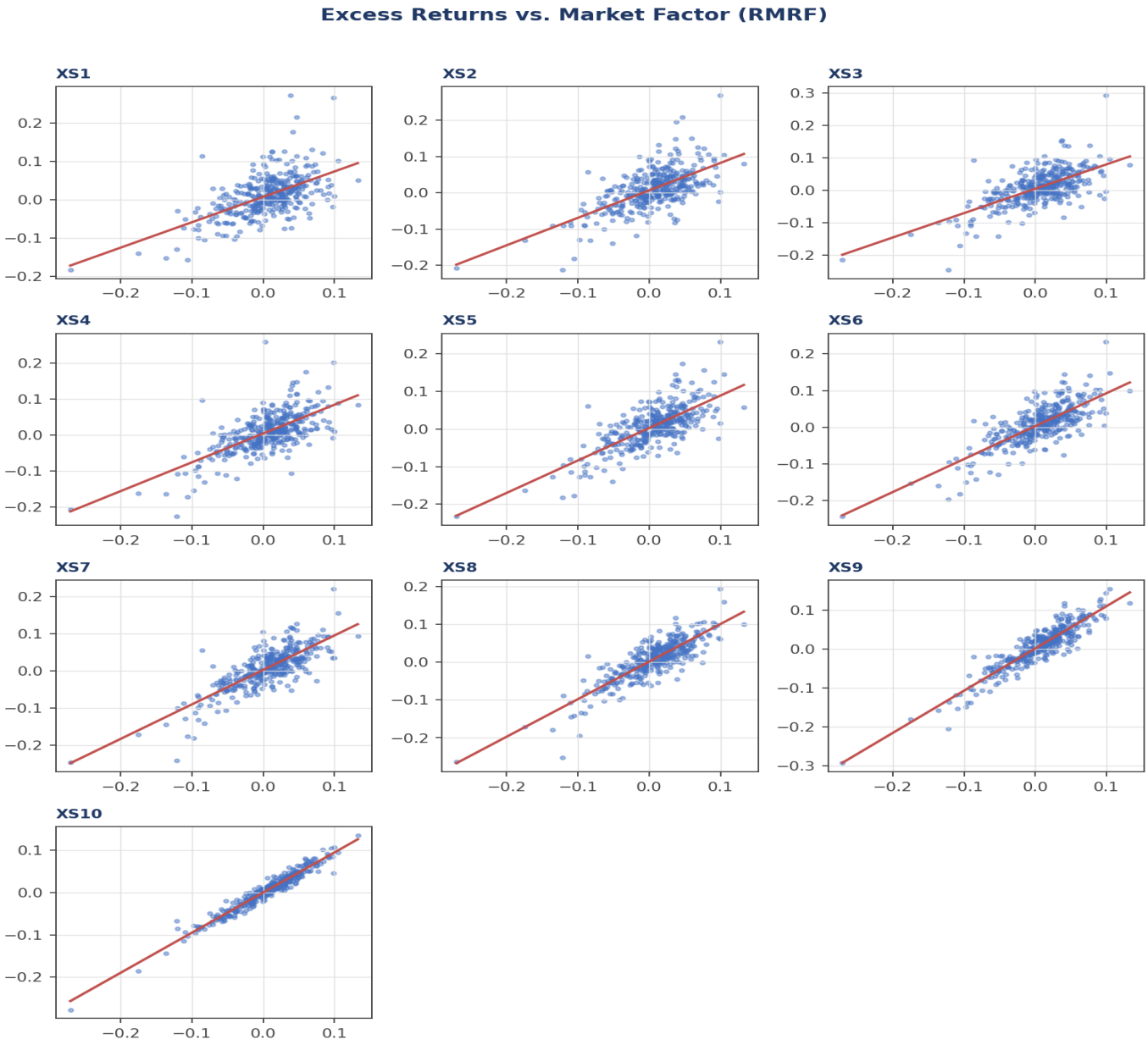


Figure 6. Excess returns vs. market factor (RMRF), with fitted regression line

As shown in Figure 6, all the portfolios move in the same direction as the market factor (RMRF). For the small portfolios the points are more scattered, meaning they are more volatile, while the larger portfolios look tighter and follow the line more closely. This tells us that bigger portfolios react more strongly and more consistently to the market, and the market factor explains their returns quite well. As for the Alpha values in Table 2, these tend to be close to 0, showing how returns are largely explained by the model, leaving only a small part to any "unexplained return."

T-Statistics

XS	T Alpha	T RMRF	T SMB	T HML	T UMD
XS1	2.83	17.12	15.57	1.87	1.88
XS2	3.48	22.26	19.52	1.16	-0.68
XS3	2.52	25.73	23.41	1.78	-0.06
XS4	1.89	30.82	28	-2.85	2.55
XS5	1.01	35.44	27.09	1.17	1.7
XS6	1.78	42.57	30.72	0.69	0.53
XS7	1.1	48.65	32.49	-0.59	0.35
XS8	0.57	58.59	31.16	-1.42	2.07

XS	T Alpha	T RMRF	T SMB	T HML	T UMD
XS9	1.36	60.74	18.61	-0.36	-2.49
XS10	-0.04	95.09	-10.91	-0.56	3.58

Table 3. T-statistics of the time-series regressions

Table 3 shows that the market factor (RMRF) is strongly significant for every portfolio, with values well above the usual cutoff of 2. The size factor (SMB) is significant for all portfolios, but while XS1–XS9 show strong positive exposure, XS10 exhibits a strongly significant negative coefficient, consistent with its large-cap profile. For the value factor (HML), the results are much weaker — most portfolios have t-values below 2. Momentum (UMD) gives mixed signals: significant for a handful of portfolios, while the others do not show a clear momentum effect.

P-Values

XS	P Alpha	P RMRF	P SMB	P HML	P UMD
XS1	0.0048	<0.001	<0.001	0.0621	0.0613
XS2	<0.001	<0.001	<0.001	0.2458	0.4944
XS3	0.0121	<0.001	<0.001	0.0763	0.9491
XS4	0.0601	<0.001	<0.001	0.0047	0.0112
XS5	0.3148	<0.001	<0.001	0.2413	0.0894
XS6	0.0753	<0.001	<0.001	0.492	0.5967
XS7	0.2701	<0.001	<0.001	0.5533	0.7235
XS8	0.5669	<0.001	<0.001	0.1577	0.0391
XS9	0.1747	<0.001	<0.001	0.7181	0.0131
XS10	0.9705	<0.001	<0.001	0.5731	<0.001

Table 4. P-values of the time-series regressions

RMRF and SMB p-values are extremely low across all portfolios, confirming high significance. HML p-values are consistently above 0.05, indicating statistical insignificance. UMD is significant for a few portfolios only. The statistical significance of the Alphas is rather unusual, as only portfolios with smaller companies (XS1–XS3) are statistically significant.

R-Squared Values

XS	R^2	Adj. R^2	F-statistic
XS1	0.6023	0.5979	135.5
XS2	0.7195	0.7163	229.5
XS3	0.7785	0.776	314.5
XS4	0.8298	0.8279	436.3
XS5	0.8496	0.8479	505.6
XS6	0.8876	0.8863	706.8
XS7	0.9073	0.9063	876.4
XS8	0.9259	0.925	1117.5
XS9	0.9218	0.9209	1055.3
XS10	0.9629	0.9625	2321.7

Table 5. Goodness of fit of the time-series regressions

R^2 values rise gradually from XS1 to XS10. The explanatory power increases as the size of the portfolio increases, which is consistent with the theoretical model, as smaller portfolios contain more idiosyncratic volatility, making them harder to explain, while larger portfolios show higher R^2 due to more stable and diversified returns.

4. Cross-Sectional Regression Analysis

4.1 Introduction

This section presents the results of the cross-sectional regressions, where the mean excess returns of portfolios XS1–XS10 are used as the dependent variable and the estimated factor betas from the time-series regressions serve as independent variables. Three separate Fama–MacBeth-style regressions are conducted, to evaluate whether the factor exposures obtained from the time-series estimations can explain the cross-sectional variation in average returns across the ten portfolios.

4.2 RMRF + SMB Regression

Regression Results

- $R^2 = 0.754$, high for a cross-sectional regression with only ten portfolios.
- $\gamma_1 \beta RMRF = -0.01026$, statistically significant ($p = 0.014$).
- $\gamma_2 \beta SMB = 0.00123$, not statistically significant ($p = 0.32$).

Interpretation

The βRMRF coefficient is the only significant factor. Its negative sign suggests that portfolios with higher market risk earned lower average excess returns during the sample period—opposite to CAPM predictions. SMB is insignificant, implying that size effects do not explain cross-sectional differences in average returns across XS1–XS10.

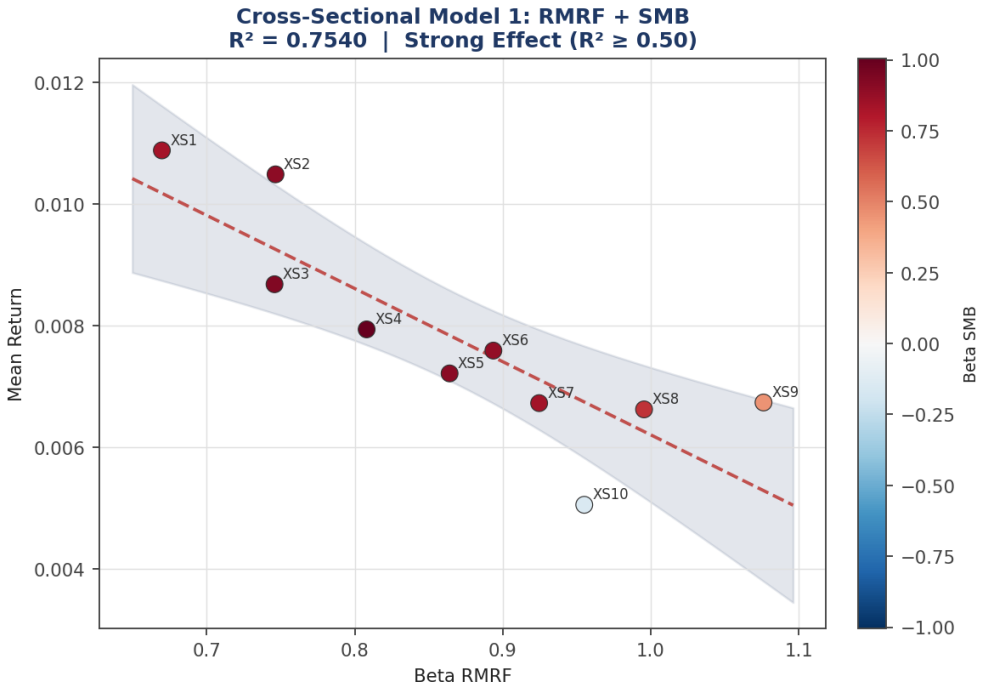


Figure 7. Cross-sectional model 1: Mean Return = $\gamma_o + \gamma_1 \beta RMRF + \gamma_2 \beta SMB$

4.3 RMRF + HML Regression

Regression Results

- $R^2 = 0.740$, indicating a relatively high fit for a cross-section with only ten observations.
- $\gamma_1 \beta RMRF = -0.01054$, weakly significant ($p = 0.015$).
- $\gamma_2 \beta HML = 0.00571$, not statistically significant ($p = 0.43$).

Interpretation

βRMRF again emerges as the main driver of returns, although its sign remains negative. The HML coefficient is positive but insignificant, indicating that value characteristics do not meaningfully differentiate returns in this sample.

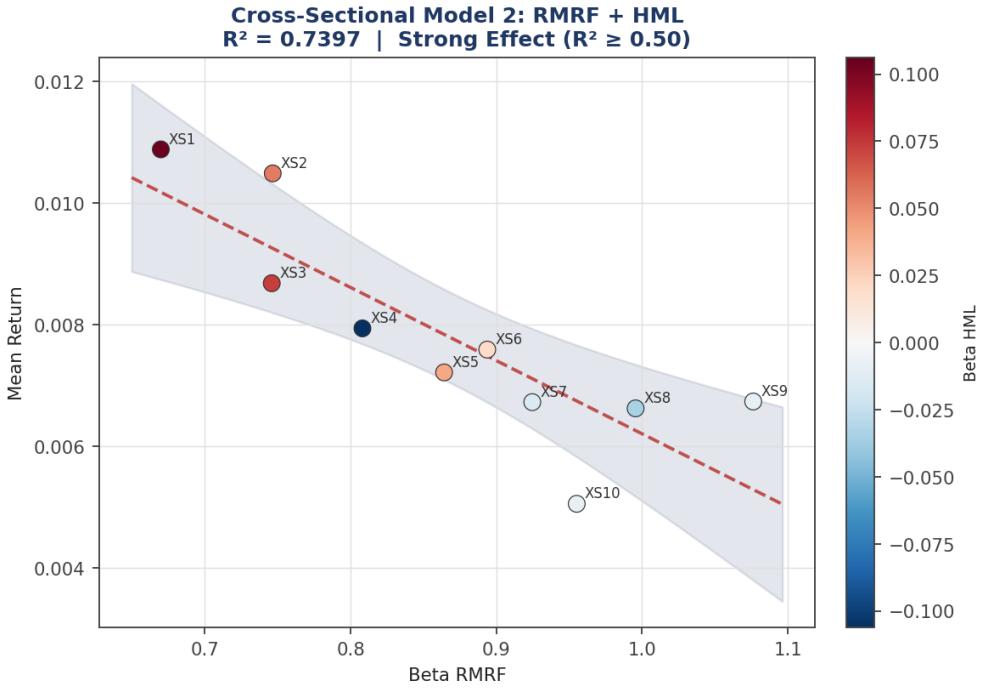


Figure 8. Cross-sectional model 2: Mean Return = $\gamma_o + \gamma_1 \beta RMRF + \gamma_2 \beta HML$

4.4 RMRF + UMD Regression

Regression Results

- $R^2 = 0.789$, the highest among the three regressions, although R^2 should be interpreted with caution given the small cross-sectional sample (only ten portfolios).
- $\gamma_1 \beta RMRF = -0.01368$, statistically significant ($p = 0.001$).
- $\gamma_2 \beta UMD = -0.01098$, not statistically significant ($p = 0.16$).

Interpretation

βRMRF stays negative and significant, while UMD is negative but not significant. Thus, momentum characteristics do not provide meaningful cross-sectional explanatory power.

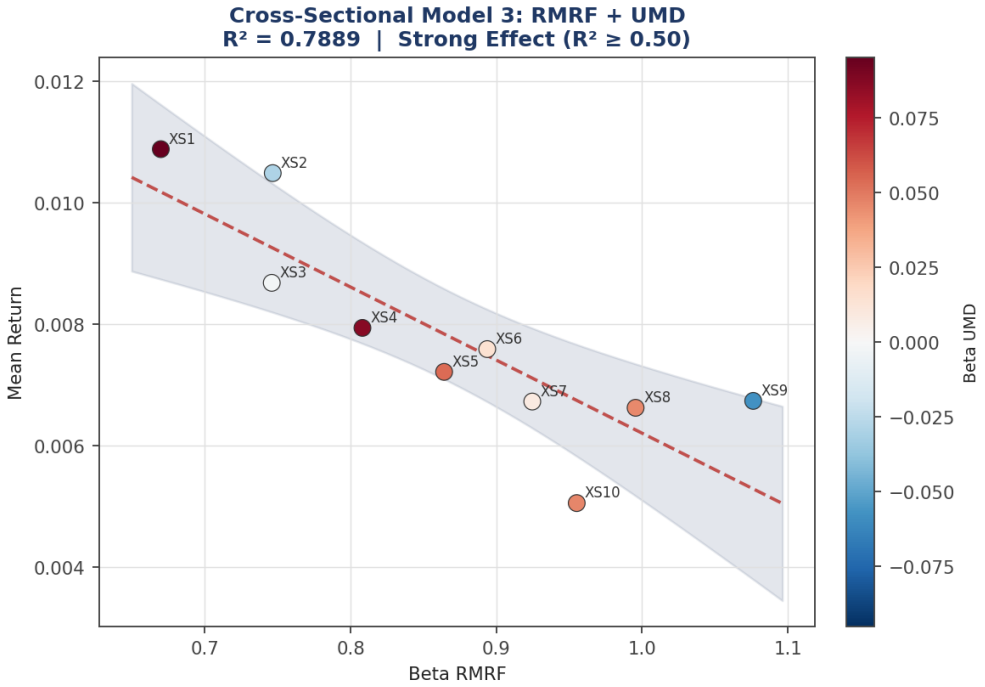


Figure 9. Cross-sectional model 3: Mean Return = $\gamma_0 + \gamma_1 \beta RMRF + \gamma_2 \beta UMD$

4.5 Cross-Sectional Comparison Across the Portfolios

The purpose of the cross-sectional regressions is to see whether the betas estimated in the time-series regressions can explain the differences in average returns across the XS1–XS10 portfolios. We expected to find some pricing impact from size, value, or momentum, but this did not appear clearly in the results: almost all of the explanatory power came from the market factor (RMRF), while SMB, HML, and UMD were statistically weak. Although the regressions showed relatively high R^2 values (0.75–0.79), these must be interpreted carefully because the cross-sectional sample includes only ten portfolios.

According to asset-pricing theory, size, value, and momentum should help explain return differences. However, this was not the case in our sample: SMB and HML were not priced, and UMD did not show a clear pattern. A notable finding is that the market price of risk (γ_1) is negative across all three models, which goes against CAPM and most multifactor evidence. This negative sign remains even after winsorisation (see Appendix), showing that it is not caused by outliers but by the nature of the dataset and the specific period analyzed.

Because the time-series regressions produced stable and reasonable betas, we believe that the weak cross-sectional results are due to the dataset itself. The portfolios appear too similar in their HML and UMD exposures, and although SMB betas differ by size, the size premium seems small in this particular sample. Overall, most of the differences in average returns across the portfolios appear to be related mainly to market exposure, rather than size, value, or momentum characteristics.

Appendix: Winsorizing Robustness Analysis

To see if outliers could have a major influence on our dataset, we decided to apply the Winsorising technique. Initially, we applied 99% Winsorising (top/bottom 0.5% — the standard threshold in academic research), but we noticed that many outliers were not eliminated. So we also implemented a more restrictive Winsorising at 95% (top/bottom 2.5%) to see how our dataset behaved. Below are some comparative elements of the three datasets to show the difference.

First, comparing the betas: the less the data is winsorised, the larger the beta values become in absolute terms — this also applies to negative values, which become more negative in the original data. Although the difference seems pronounced in the graphs, the scale of values is narrow, so the data does not change much; the influence of outliers appears limited, as beta estimates remain largely stable under 99% winsorisation. With 99% winsorised data, results are almost identical to the originals: extreme outliers (top/bottom 0.5%) do not significantly influence the beta estimate. With 95% winsorised data, the differences are more pronounced — the RMRF betas tend to flatten towards the bottom (for example, the RMRF beta of XS1 drops from 0.67 to 0.57) — but the overall structure (increasing trend for RMRF, decreasing trend for SMB) remains intact.

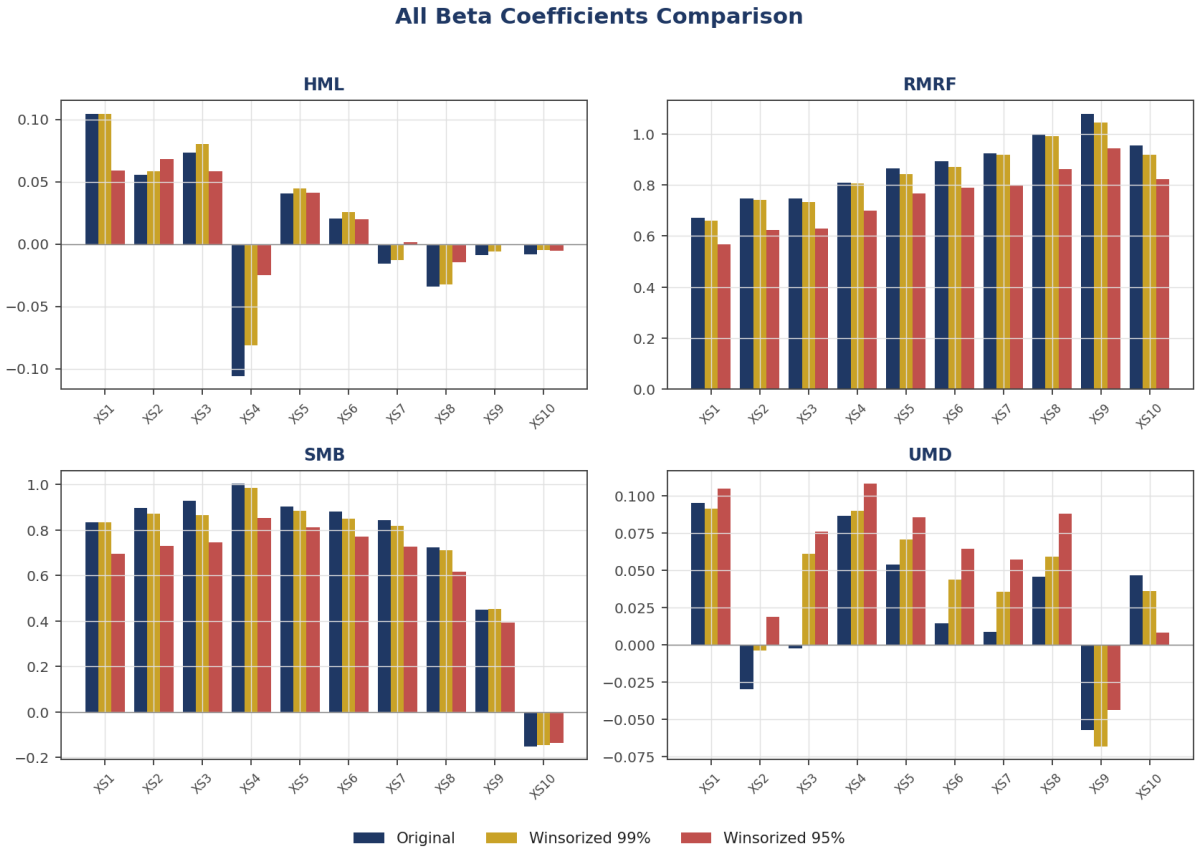


Figure 10. Beta coefficients across the original, 99%-winsorised and 95%-winsorised datasets

Comparing the R^2 values also indicates a reasonably strong fit, consistent with multifactor asset-pricing regressions. The values for the original and 99% winsorised data are almost identical, while at 95% the fit loses a few percentage points and the effectiveness of the model gradually decreases. By removing extreme variability, the factors are able to explain a smaller portion of the differences in returns between portfolios.

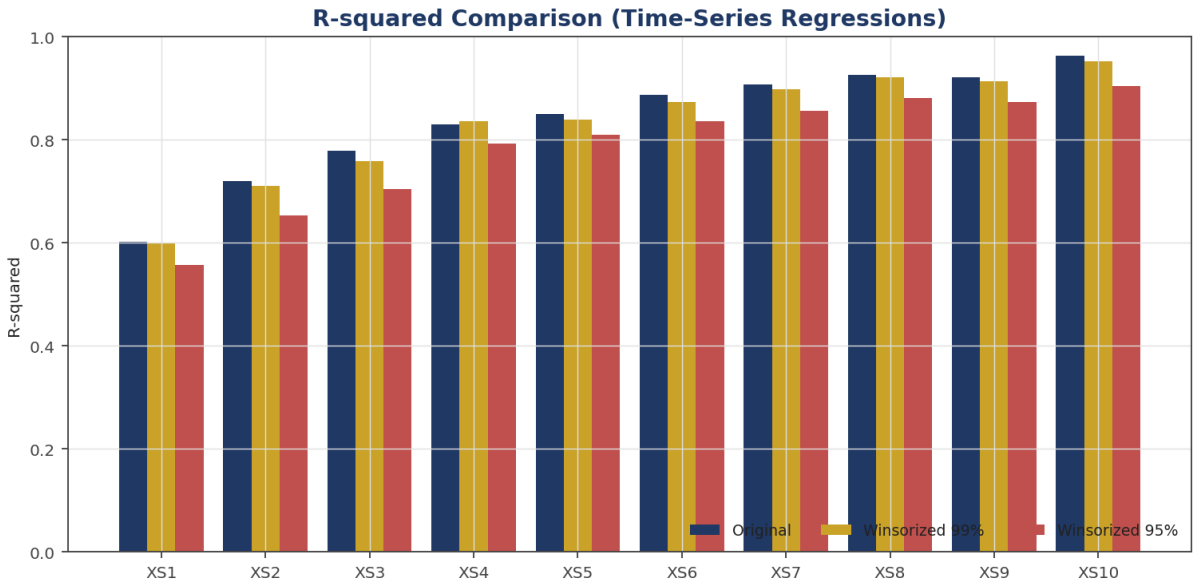


Figure 11. R^2 of the time-series regressions across the original, 99%-winsorised and 95%-winsorised datasets

Impact on Cross-Sectional Analysis

Dataset	Model 1 (RMRF+SMB)	Model 2 (RMRF+HML)	Model 3 (RMRF+UMD)
Original	0.7540	0.7397	0.7889
Winsorised 99%	0.7113	0.6873	0.7755
Winsorised 95%	0.6545	0.6514	0.6256

Table 6. Cross-sectional R^2 across the original, 99%-winsorised and 95%-winsorised datasets

Even with 95% of the data cleaned, the market price of risk (γ_1 on βRMRF) remains negative in every specification. This confirms that the negative sign is not driven by a handful of extreme data points, but is a structural feature of the sample analysed. As Table 6 shows, R^2 values follow the same pattern observed in the time-series regressions: higher in the original data and decreasing progressively with stronger winsorising.

Conclusions

The combined analysis of descriptive statistics, time-series regressions, cross-sectional regressions, and the main statistical tests provides a comprehensive view of the risk–return structure of the UK portfolios. The time-series regressions clearly show that most of the excess returns are strongly associated with the market factor, with positive market betas and SMB patterns that align with portfolio size (higher SMB exposure for small portfolios and lower exposure for large ones).

On the cross-sectional side, the relationship between average returns and market betas offers insights into the actual pricing of systematic risk. The inclusion of SMB, HML, and UMD was intended to capture additional dimensions of risk, but these factors turned out to have limited pricing power in this particular dataset.

Finally, from a Winsorisation perspective, the distribution of returns is clearly non-normal. The robustness checks confirm that aggressive data cleaning (95% Winsorisation) is counterproductive, as it removes meaningful variation and lowers the explanatory power of the model. Mild trimming (99% Winsorisation) produces results nearly identical to the original dataset, showing that the influence of outliers is minimal. Therefore, the original data appear to be the most appropriate and informative for our analysis.

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